

Predicting Recommendation Probability Using Regression Model on User Interaction on E-commerce Product

Transforming Raw Data into Actionable Insights



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Project Overview

- Create a model for an e-commerce platform
- Accurately predict how likely a product is to be recommended to customers.
- Improve the relevance of product suggestions.
- Drive higher user interaction and sales.
- Ultimately lead to happier customers through better recommendations.



Objectives



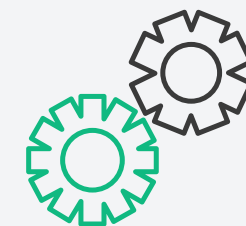
**Develop and Evaluate a
Content-Based
Recommendation Model**



**Identify Key
Features
Influencing Product
Recommendability**



**Analyze Results and
Generate Useful
Insights**



**Address Potential
Biases**



Dataset Description

Here are the features of our dataset:

- Number of clicks on similar products
- Number of similar products purchased so far
- Average rating given to similar products
- Gender
- Median purchasing price
- Rating of the product
- Brand of the product
- Customer review sentiment score (overall)
- Price of the product
- Holiday
- Season
- Geographical locations
- Probability for the product to be recommended to the person (target variable)

Data Preparation



- Cleaned the data via checking no values
- Removed duplicates
- Converted text data into numerical representation
- Applied scaling of data (to be in the same range)
- Removed one-third of the columns that has minimal effect on outcome for better prediction



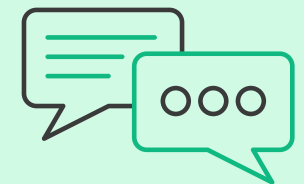
Model Selection →

RANDOM FOREST REGRESSOR



Think of it like this:

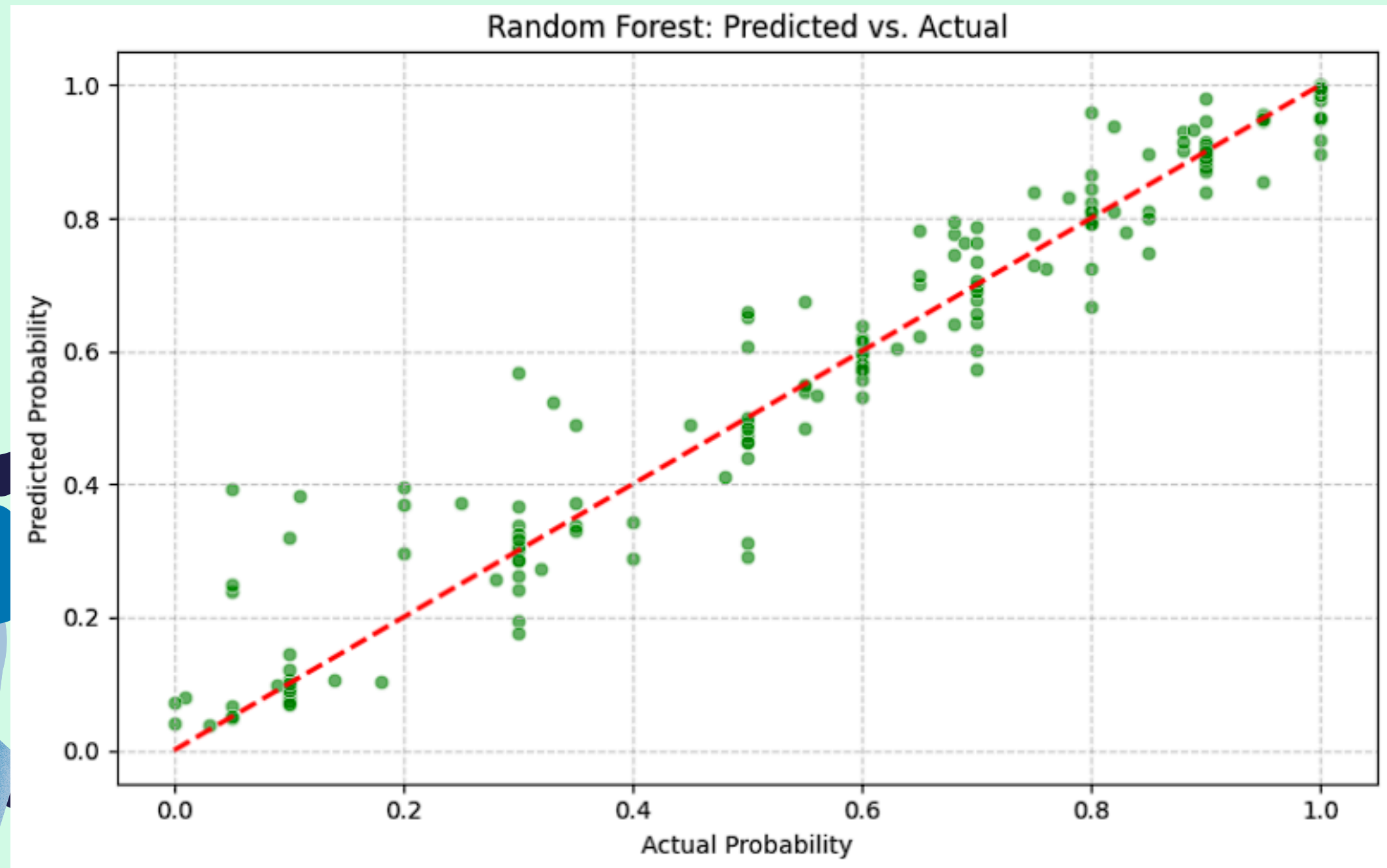
Imagine you're trying to forecast the stock market or predict customer behavior. Instead of relying on a single analyst's opinion, which might be biased or miss important nuances, a Random Forest Regressor harnesses the wisdom of a diverse team of experts.



Graphical Result

As we can see, the data points are clustered around our model.

This means that the prediction of the model is close to actual value.



Analytical Result



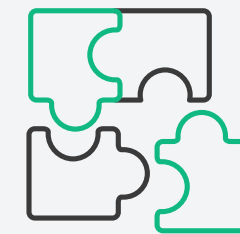
These numbers shows the percentages of the model's prediction in comparison to the actual value.

We can see that it is highly accurate with tolerable margin of wrong predictions



Mean Absolute Error

4.34%



Mean Squared Error

0.52%



Root Mean Squared Error

7.22%

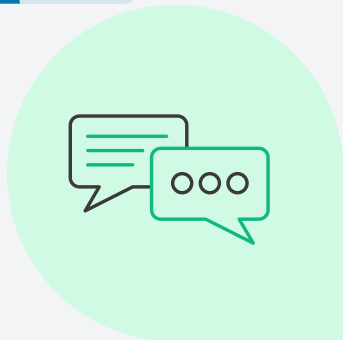


R-Squared (Accuracy)

94.36%

Insightful Observations

Here are the **top** 5 features that impact recommendability of the product.



RANK	FEATURE	IMPORTANCE
1	Customer review sentiment score (overall)	64.01%
2	Rating of the product	19.86%
3	Average rating given to similar products	6.81%
4	Price of the product	3.15%
5	Number of similar products purchased so far	1.74%

Insightful Observations

Here are the **least** 5 features that is **not** affecting recommendability of the product that much.

All of those are of feature "brand."



RANK	FEATURE (Brands)	IMPORTANCE
1	Lijoba	0.0021%
2	Dabur Chyawanprash	0.0025%
3	Levis Strauss & Co.	0.0028%
4	Lee Cooper	0.0029%
5	Spyker	0.0056%

Interpretation of Results



- Less impactful features, particularly some specific brand categories, suggest that their individual presence might not strongly dictate recommendation probability.

- Customer Review Sentiment and Product Rating consistently emerged as the most critical features across both models, underscoring the importance of product quality and user satisfaction in driving recommendations.



Bias & Fairness Analysis

While **relying on customer reviews and ratings** is generally sound, **their subjectivity** and **potential for manipulation** warrant careful consideration.

furthermore, **sensitive attributes** such as **gender, brands, and geographical locations** **require rigorous auditing** to ensure **fair recommendations across all** demographic groups.



Thank
You

